

BACHELOR THESIS · COLLOQUIUM

Identification and Interpretation of BMS Signals Through CAN Bus Reverse Engineering on the Volkswagen ID. Buzz

Muhamad Alif Izzuwan Bin Ibrahim

Electrical Engineering and Electromobility · Technische Hochschule Ingolstadt

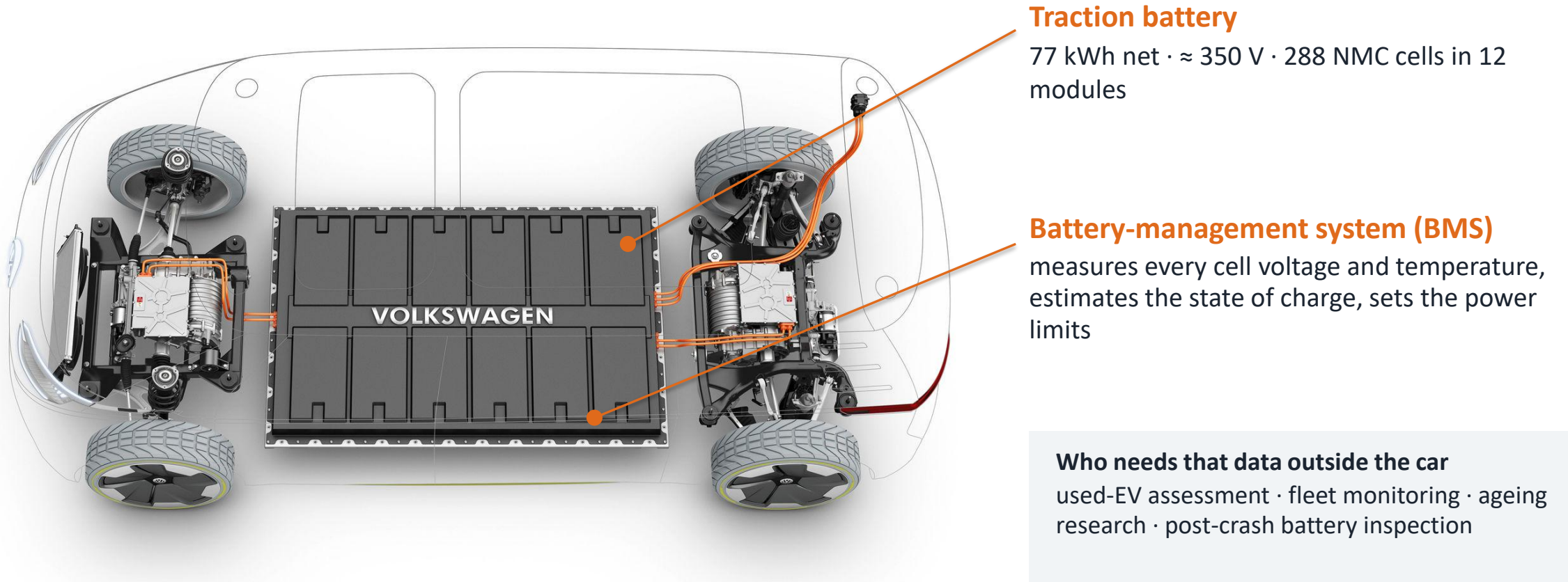
Examiners: Prof. Dr. Hans-Georg Schweiger · Prof. Dr. Werner Huber

Supervisor: M.Sc. Markus Gregor

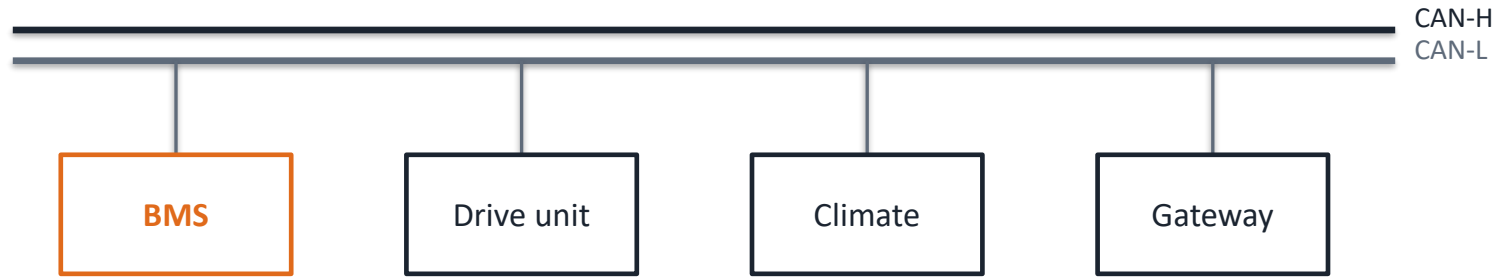


Photo: Volkswagen AG

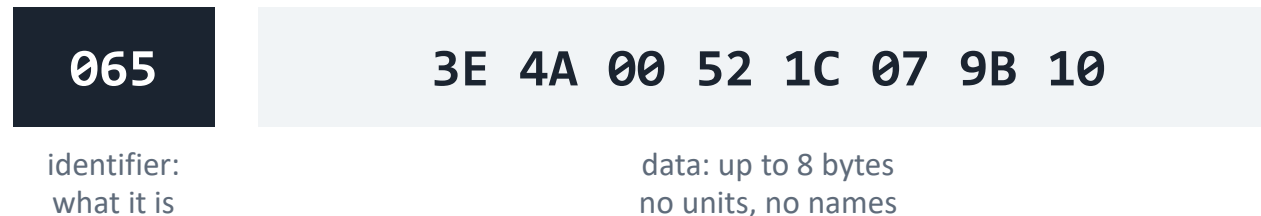
The battery computes a detailed picture of its own state, and almost none of it leaves the car



Inside the car, control units share one two-wire line, and every message is only a number plus up to 8 bytes



One frame on the bus, as the logger records it



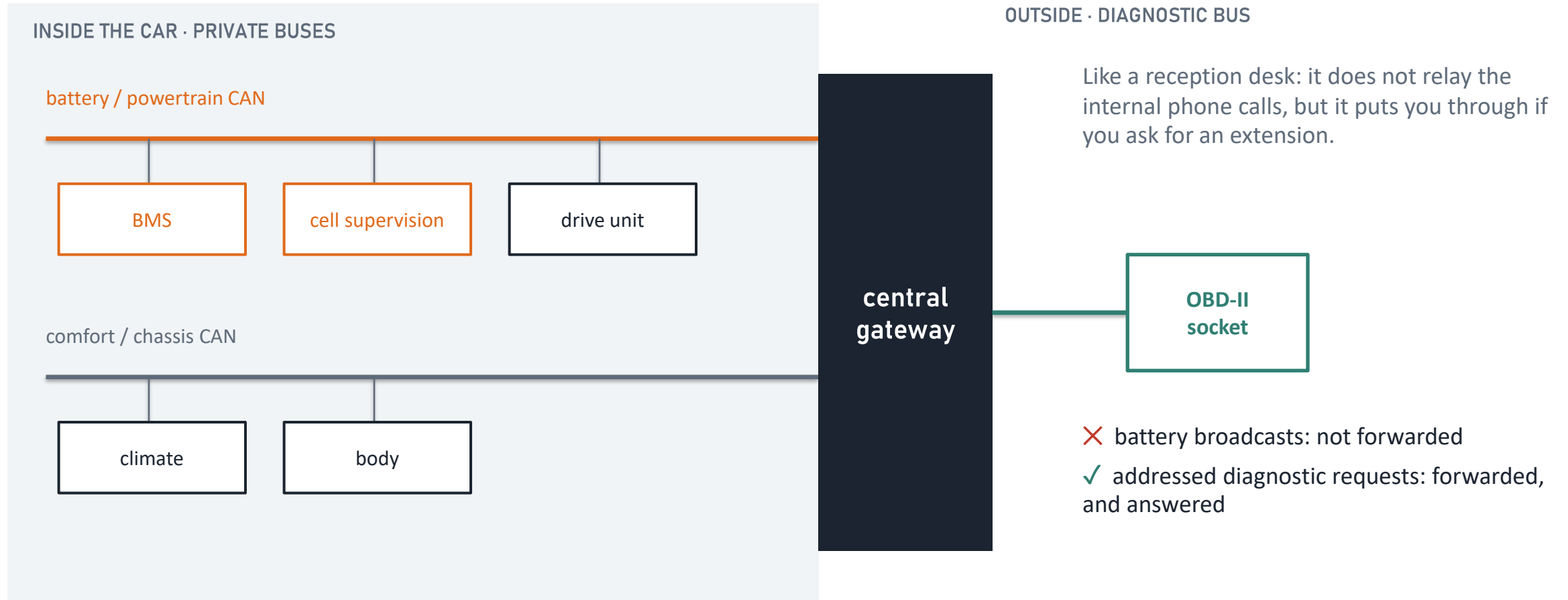
Think of a radio channel

Every unit hears every message. A message carries a topic number, not an address.

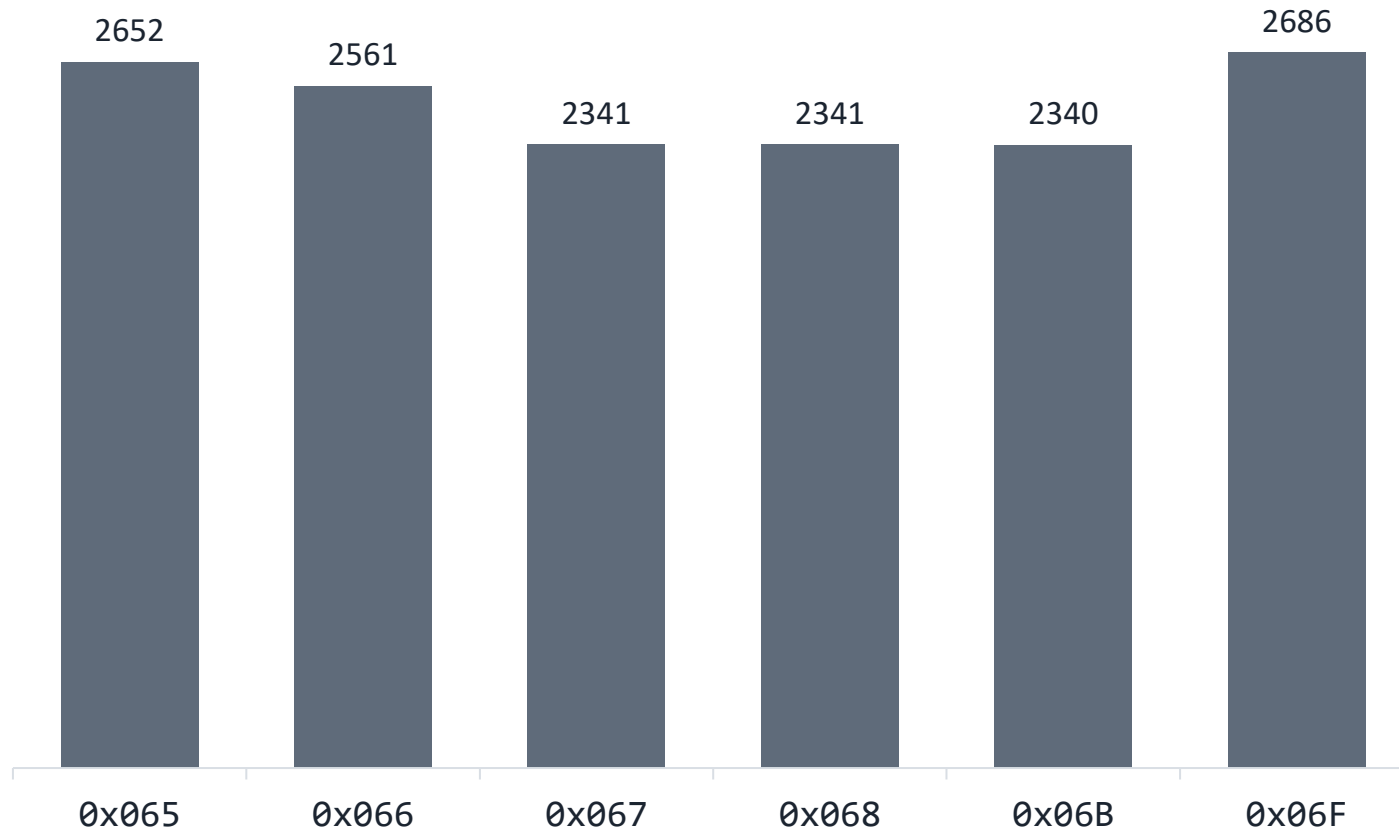
The bytes are visible.

Their meaning is not: the dictionary that maps bytes to volts and degrees belongs to the manufacturer.

On the MEB platform a gateway keeps the battery's bus away from the diagnostic socket



Listening at the socket for 533 s recorded 14,921 frames, and every one came from the logger itself, none from the car



Unsolicited frames per 11-bit identifier, log 00000039

14,921

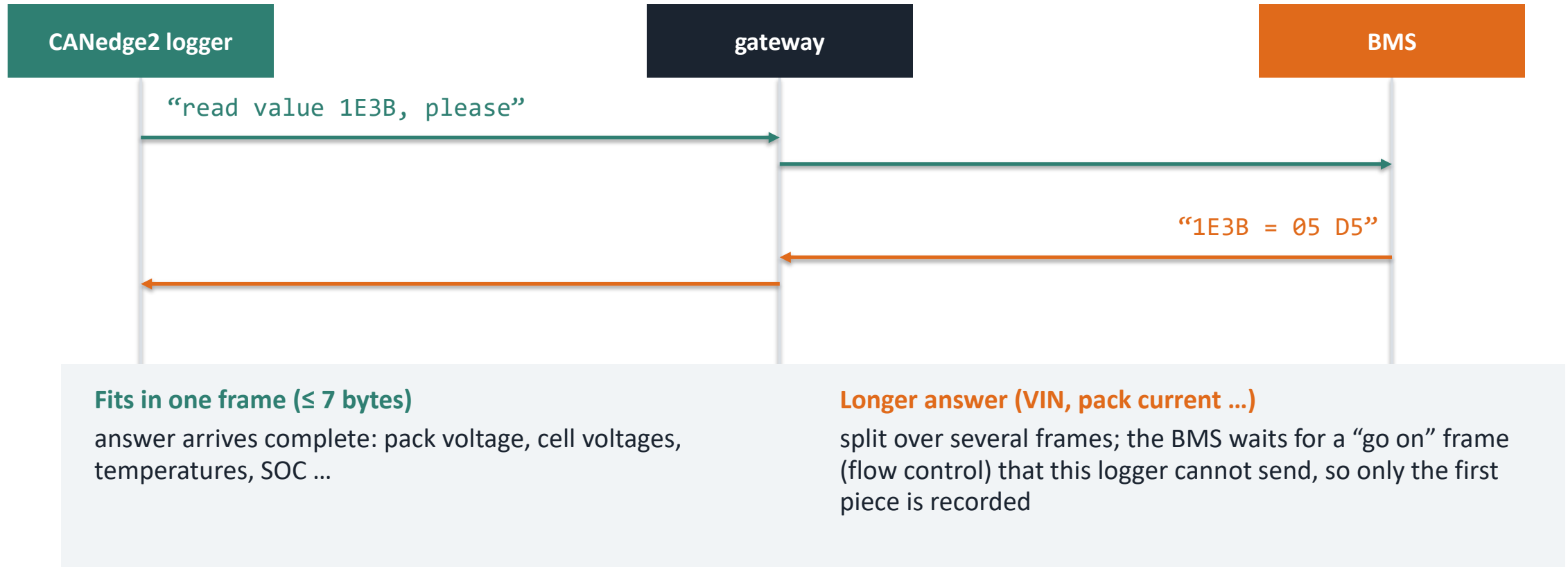
unsolicited frames on 0x065 – 0x06F: the logger's own GNSS/IMU messages

0

broadcast frames from the car, and none from the battery

Passive reverse engineering has nothing to work on here.

So the logger asks instead: one question frame out, one answer frame back



ReadDataByIdentifier, service 0x22 of ISO 14229 (UDS), carried by ISO 15765-2 (ISO-TP)

MATERIALS

One logger on the OBD-II socket asks, listens and records on one clock, with no laptop on the bus



CANedge2: CAN logger with SD card, can transmit a fixed request list

OBD-II → DB9 cable: CAN-H, CAN-L, GND, 12 V

500 kbit/s

Classical CAN

60 identifiers

per request cycle

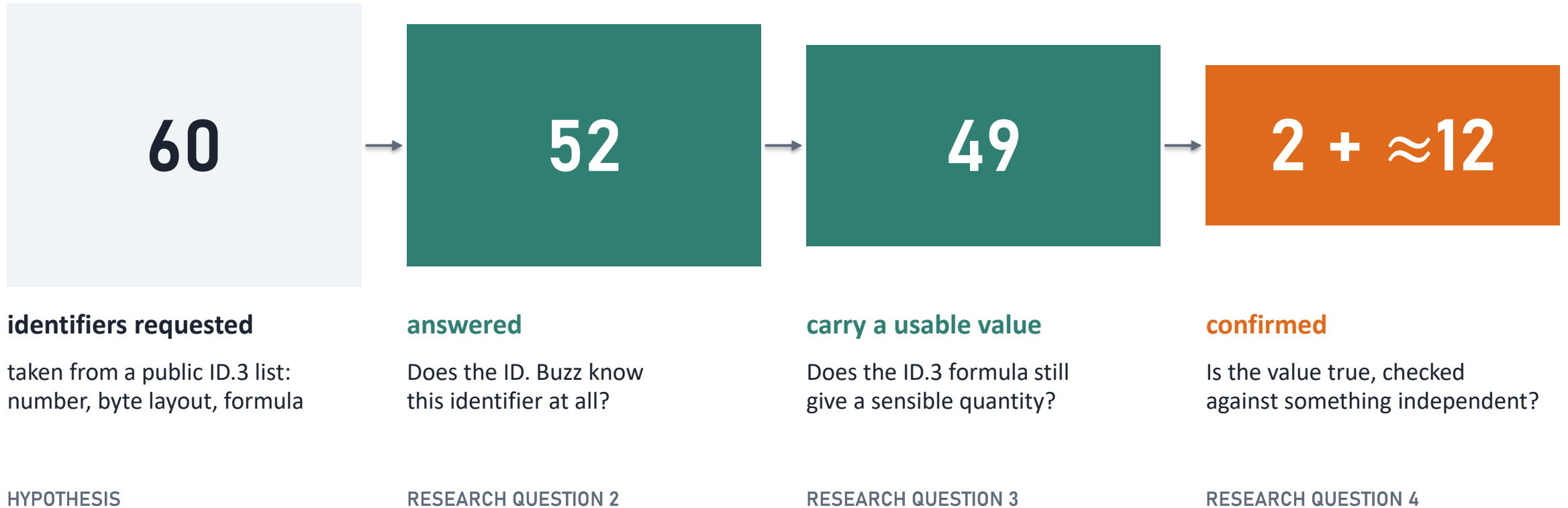
10 s cycle

requests 150 ms apart

3 sessions

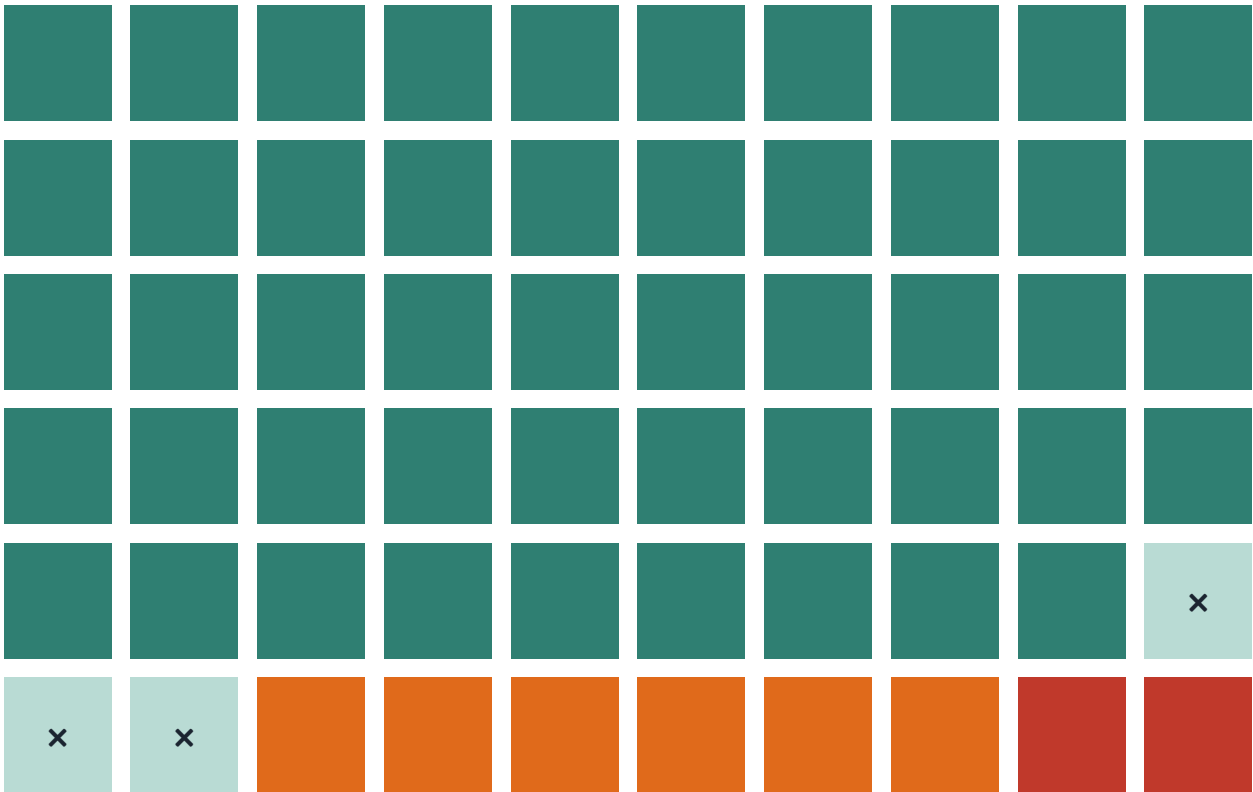
identical outcome

A list written for the VW ID.3 supplied the candidates; every entry had to pass three tests on the ID. Buzz



The list is community-compiled, unvalidated and made for a different car. Treating it as a specification would have reproduced its errors.

52 of 60 identifiers answered, and every failure traces back to the measurement setup, not to the car



- 49 answered with a usable value**
47 from the BMS, 2 from drive unit and climate
- 3 answered “not available”**
code 0x0FFE: an empty cell slot, explained on slide 12
- 6 only the first piece arrived**
answer > 7 bytes; the logger cannot ask for the rest
- 2 no answer at all**
gateway identifiers, queried without a diagnostic session

One square = one identifier · outcome identical in all three sessions

One answer, decoded byte by byte, gives the voltage of the whole battery pack: 373.25 V

QUESTION

03

3 bytes follow

22

read data

1E 3B

which value

55 55 55 55

padding

ANSWER

05

5 bytes follow

62

positive reply

1E 3B

echo

05 D5

the value

aa aa

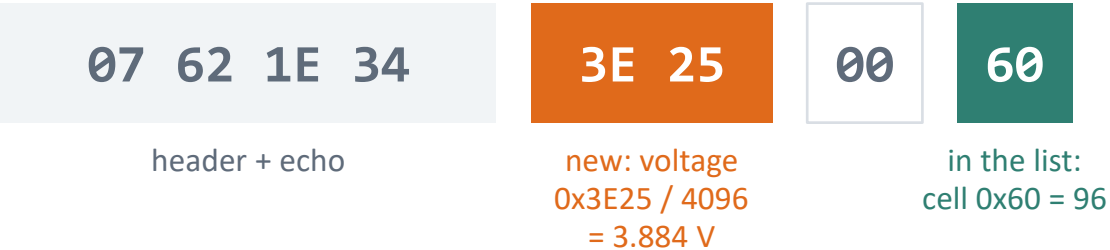
unused

$0x05D5 = 1493 \rightarrow 1493 / 4 = 373.25 \text{ V}$

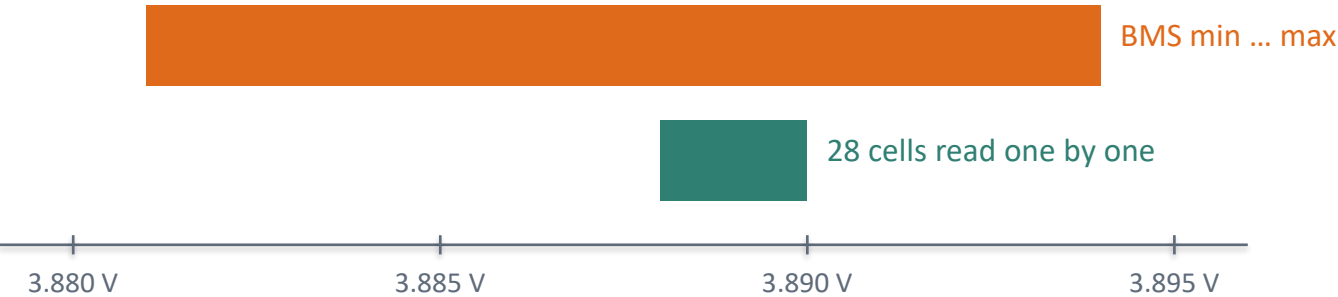
SAME SESSION, OTHER VALUES	
State of charge	72.8 – 73.2 %
Pack voltage	373.25 – 373.75 V
Cell voltage (28 sampled)	3.888 – 3.890 V
Battery temperature	16.0 – 16.4 °C
Discharge limit	800 A
Operation mode	drive-ready
Speed	0 km/h

The highest- and lowest-cell identifiers carry a voltage the ID.3 list never mentions, scaled by 1/4096

ANSWER OF 0x1E34, “LOWEST CELL”



WHY 1/4096 IS RIGHT: THE EXTREMES MUST ENCLOSE EVERY SAMPLED CELL



112 / 112

reads across three sessions name cell 96 as the lowest cell.

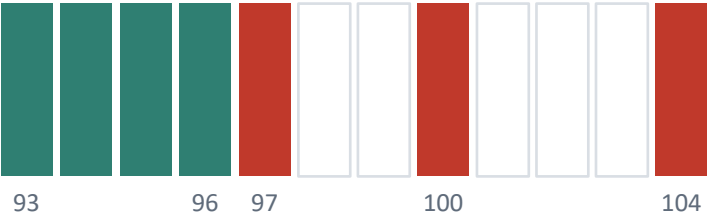
The highest cell wanders with the noise; the lowest does not.

Lesson learned

A first decoding read the wrong byte and reported “cell 62” as both highest and lowest: impossible, and that is how it was caught.

0x0FFE is not a 5.09 V cell but the code for “no cell here”, and three arguments say the pack measures exactly 96 cells

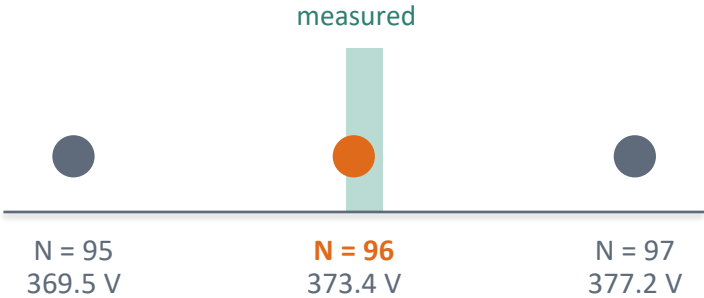
0x0FFE → “5.094 V”



teal = cell present · red = 0x0FFE · white = not sampled

96

cell-voltage channels



1 From above

slots 97, 100 and 104 return 0x0FFE in 100 % of reads: the largest 12-bit value minus one, a reserved “not available” code

2 From below

the BMS itself names cell 96 as its lowest cell in 112 of 112 reads, so cell 96 exists

3 From the pack voltage

373.5 V / 3.8891 V per cell = 96.0; any other count misses the measured band

Volkswagen publishes 288 cells in 12 modules; 96 series groups means three cells in parallel per group.

Only two signals are confirmed against the real world; the rest rest on physics or on plausibility

A	directly validated 2 signals	operation mode = drive-ready, speed = 0 km/h: matches what was observed in the car
B	indirectly validated ≈ 12 signals	pack voltage, cell voltages, cell extremes, temperatures: tied by relations no formula was fitted to ($V_{\text{pack}} / V_{\text{cell}} = 96.0$, extremes enclose samples)
C	plausible, unconfirmed the remainder	state of charge, power limits, pack current, 12 V, outdoor temperature, odometer: sensible magnitude, no independent reference
D	invalid or unexplained 6 signals	three 0x0FFE slots, A/C compressor status, two unanswered

The method works at a standstill; current, charging and battery health are the next step

LIMIT IN THIS WORK

WHAT REMOVES IT

All recordings stationary: current and power never left idle

→ Drive at 50 and 100 km/h, accelerate, brake, DC fast charge

Logger cannot send flow control: 6 long answers truncated, incl. pack current

→ Read them with an ISO-TP-capable tester

Gateway queried without a diagnostic session: 2 silent identifiers

→ Open a session on the gateway, or address the right unit

State of health not requested; protected identifiers need security access

→ Only with manufacturer cooperation, not by bypassing it

Application: after a crash, SOC, cell spread and temperatures read non-invasively through the socket.

Asking instead of listening reaches the battery through a production gateway

RQ 1 The socket carries the diagnostic bus; the gateway keeps the battery's broadcasts private. Not one broadcast frame of the car reached the socket.

RQ 2 Without unlocking: 52 of 60 identifiers answer, identically in every session; all 8 failures trace back to the logger or the session setup.

RQ 3 The ID.3 formulas mostly transfer; three things they lack were found: the 1/4096 cell extremes, the 0x0FFE code, the 96-cell count.

RQ 4 2 signals validated against the real world, ≈ 12 by physical relations; the rest are plausible and are reported as such.

Thank you. I look forward to your questions.